

# Gideon A. Afriyie

AI RESEARCH · DATA ANALYSIS · QUANTITATIVE RESEARCH

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## Experience & Research

### Google

May 2026 - Present

TRC RESEARCH 🔗

- Selected and awarded Google Cloud TPU infrastructure grant for ML research on extreme statistical models.
- Research findings expected for publication; Google TRC participants required to share results publicly.

### Portland State University

June 2026 - August 2026

ALTREU RESEARCH (TEAM OF 4) 🔗

Remote | Part-time

- Built ML pipeline predicting LA daily AQI from 10 years of NASA and EPA data; model achieved  $R^2$  of 0.825, beating target baseline by 44.6%.
- Engineered a 3,650-day, 5-pollutant AQI dataset with zero missing values from EPA AQS API.

### MIT/Harvard IAIFI

August 2026

AI SUMMER SCHOOL 🔗

- Attended lectures on symbolic regression, diffusion models, and simulation-based inference.
- Collaborated in a hackathon applying Physics-Informed AI methods with hands-on tutorials.

### Gettysburg College

Aug 2026 – Present

3D TOPOGRAPHIC MODELING R.A

On-campus

- Developing 3D terrain models from regional topographic and elevation data for K-12 STEM education.
- Researching digital modeling and fabrication workflows, building prototypes, and refining models for classroom use.

### Under Canvas Inc.

May 2025 - July 2025

DATA ANALYTICS FELLOW

- Analyzed a \$1.6M dataset using Python/Excel to model bonus inconsistencies and cost efficiency.
- Proposed a data-driven incentive framework to optimize corporate cost efficiency.

## Publications

Afriyie, G. A. “Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér’s Conjecture and Light-Tailed Behavior.” *Journal of Statistics & Probability Letters* (Elsevier). DOI: 10.1016/j.spl.2026.110898

## Honors & Awards

- 2026 **AMS - Math Alliance F-GAP Scholar**, a selective national program recognizing mathematical talent
- 2026 **Taylor Gaw ’13 Endowed Fellowship**, competitive institutional fellowship
- 2024 **Seibert Project Award**, recognized for leadership, mentorship, and community impact
- 2023 **David Wills Scholar**, merit-based academic excellence award (\$33,000/yr) - 1530 SAT (780/800 Math)

## Education

Gettysburg College, B.A. Mathematics, Economics minor

2023 - 2027

Gettysburg, PA

BAUM Tenpers Institute, SDEs, Stochastic Calculus, Dynamic Programming

Summer 2025

Grad-Coursework

## Skills

**Tools** Python, R, SQL, LaTeX, Excel  
**Modeling** EVT, Time-Series, Statistical Inference